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Rajshahi University of Engineering & Technology (RUET)
Department of Mechanical Engineering

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INDUSTRIAL TRAINING REPORT

Industrial Training at

Confidence Power & Energy PLC, Bogura

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Abstract

This report shows the outcomes of an academic industrial attachment completed at Confidence Power & Energy PLC, Bogura, Bangladesh. It is a heavy fuel oil (HFO) based independent power producer facility in northern Bangladesh. The crucial purpose of the attachment was to bridge the gap between undergraduate mechanical engineering theory and the operational realities of a utility-scale reciprocating-engine power plant. It is an Independent Power Producer (IPP) generating electricity for the Bangladesh Power Development Board (BPDB) under a 15-year Power Purchase Agreement (PPA). Thus, CPEPLC has a 113 MW/unit capacity with a combined-cycle configuration comprising 6 MAN 18V48/60TS medium speed engines and 2 steam-turbine generators that utilize waste heat from the engines for electricity production. During this industrial training, focused on the plant configuration, MAN 18V48/60TS, fuel handling and conditioning, lubricating-oil systems, start and instrument air, water treatment and high-/low-temperature water circuits, exhaust gas heat recovery, cogeneration, steam, power evacuation, and SCADA. It also gained insight into the operation, maintenance, and bunker/fuel-handling practices of CPEPLC. In particular, it was able to deepen understanding of the interrelation of auxiliary systems with the prime mover to ensure reliable, secure, and economic electricity generation. Similarly, my knowledge of thermodynamics, fluid mechanics, heat transfer, internal-combustion engines, power-plant engineering, machine maintenance, instrumentation, and safety was augmented during the training. Besides theoretical concepts, this attachment improved soft skills such as teamwork, communication, observational skills, discipline, and problem-solving in the host organization. Overall, this industrial training opportunity enabled me to bridge the gap between academic curriculum and practical aspects of mechanical engineering by providing hands-on experience with modern thermal-power plants and understanding the intricacies involved in uninterrupted electricity production.

Keywords: Power plant, Heavy Fuel Oil (HFO), MAN 18V48/60TS Engine,

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List of Abbreviations

Abbreviation	Meaning
AC	Alternating Current
BPDB	Bangladesh Power Development Board
PLC	Public limited company
CT	Current Transformer
DO	Diesel Oil
EGB	Exhaust Gas Boiler
FTP	Fuel Treatment/Transfer Plant
HFO	Heavy Fuel Oil
HMI	Human-Machine Interface
HPTC	High-Pressure Turbocharger
HT	High Temperature
IAC	Instrument Air Compressor
IAS	Instrument Air System
IPP	Independent Power Producer
LT	Low Temperature
LPTC	Low-Pressure Turbocharger
LO	Lubricating Oil
MAN	Maschinenfabrik Augsburg-Nürnberg
SCADA	Supervisory Control and Data Acquisition

Chapter 1: Introduction

1.1 Company Profile

Confidence Power & Energy PLC (CPEPLC) forms a part of the Confidence Group, a major diversified group in Bangladesh with interests spanning manufacturing, infrastructure services, telecommunications and energy. CPEPLC itself was established in 2017 and has grown rapidly into one of the more significant Independent Power Producer (IPP) operating in the country. At present, the company owns and operates four HFO-fired power plants located in Bogura, Rangpur and Chattogram together with an integrated oil-and-shipping operation at Natakhola. The combined generation portfolio of these plants approaches 400 MW and is supported by a fleet of MAN and Rolls-Royce medium-speed engines.

The particular facility that hosted the industrial attachment is situated at Birgram under Shajahanpur Upazila in Bogura District. The site occupies approximately 14.5 to 16 acres and is strategically positioned to serve the northern load centers of Bangladesh. Northern Bangladesh has experienced sustained growth in both industrial and residential electricity demand, while the expansion of natural-gas pipeline infrastructure into the region has lagged behind that growth. Consequently, HFO-based reciprocating-engine plants have played an important bridging role in maintaining supply reliability.



Figure 1.1: Confidence Power & Energy PLC (CPEPLC), Bogura

The Bogura Complex comprises of two similar units each with a net contract capacity of 113 MW. Each is a combined cycle plant. The main electricity-producing units are six MAN 18V48/60TS four-stroke medium-speed reciprocating engines. The waste heat from the engines is recovered in exhaust gas boilers to produce steam for the MAN steam turbines also rated at 3.6 MW each. Thus in terms of thermal efficiency, the combined cycle scheme is more efficient than a simple cycle plant of the same capacity.

The main source of fuel is Heavy Fuel Oil (HFO) known as furnace oil in Bangladesh. The oil is procured in the international market primarily from Singapore by mother ships, and then it is transferred to Chittagong Port. The lighter vessels (typically 700-1500 MT capacity) owned or chartered by Confidence Group carry oil from Chittagong to the river terminal at Natakhola in Pabna District. From Natakhola, it is transported by road tankers to the Bogura Complex. The oil is stored in a tank farm having a total capacity of about 8000 MT of HFO. The storage comprises 1×5000 MT, 2×1000 MT settling tanks, and 1×1000 MT service tanks. Another 500 KL Diesel Oil tank serves as a backup and is used for engine starting and emergency purposes.

The Letter of Intent was issued on 10 August 2017, and the Power Purchase Agreement was signed on 22 February 2018. The Commercial Operation Date (COD) for unit 2 was reached in March 2019, and for unit 1, in November 2019. Thus, each unit has a fifteen-year Power Purchase Agreement (PPA) with BPDB, and both are set to retire in 2034. The plants are subject to specific obligations regarding availability, declared capacity, and technical performance.

The main asset of the Bogura complex is electricity generated by the HFO-fueled reciprocating engines. The six MAN 18V48/60 reciprocating engines along with a steam turbine and an exhaust gas boiler allow for a combined cycle generation scheme. It can be seen that besides the main electricity-generating units, the plant has several supporting systems that make up the infrastructure for natural gas processing, power generation, transmission and distribution, storage, and transportation.The company's official website states that its IPP plants use HFO-fired generation and that its fuel-supply operation integrates furnace-oil import, inland carriage, storage, and delivery to power plants. The company also describes the Bogura projects as important contributors to dependable electricity supply in North Bengal.

The attachment presentation identifies exposure to the following functional areas:

- Operations Department: Responsible for continuous monitoring, control, and coordinated operation of generating units and auxiliaries.
- Maintenance Department: Concerned with inspection, servicing, troubleshooting, preventive maintenance, and restoration of mechanical equipment.
- Bunker/Fuel Handling Section: Associated with receiving, storing, conditioning, transferring, and supplying HFO and supporting fuels.
- Water Treatment/Utility Areas: Responsible for preparing water for plant and cooling applications and supporting reliable operation of heat-rejection and steam systems.
- Electrical and Control Areas: Associated with alternators, switchgear, synchronization, transformers, instrumentation, and SCADA-based supervision.

Table 1: Summary of key technical and contractual data of the Bogura plant

Parameter	**Basic Information**
Plant type	HFO-fired combined-cycle IPP
Net contract capacity	113 MW per unit
Engine configuration	6 × MAN 18V48/60TS (~18.55–18.6 MW)
Steam-turbine generators	2 × 3.6 MW MAN STG
Primary fuel	Heavy Fuel Oil (Furnace Oil)
Total HFO storage	≈ 8 000 MT
LOI / PPA dates	10 Aug 2017 / 22 Feb 2018
COD (Unit 2 / Unit 1)	March 2019 / November 2019
Contract duration	15 years (retirement 2034)
Site area	≈ 14.5–16 acres

The plant's significance for Bangladesh's electricity system stems from multiple factors. First, it offers much needed reliable and dispatchable capacity in a country witnessing growing power demand. Second, the plant's ability to ramp up quickly and follow the load – due to medium-speed engines – makes it valuable for regulating the system frequency and voltage, especially in the

context of renewable energy integration. Third, the vertically-integrated supply chain of the Confidence Group makes it less vulnerable to external supply-side shocks. At the same time, the plant suffers from the typical drawbacks of a residual-fuel power generation asset: compared to gas or renewables, it emits higher quantities of sulphur oxides, nitrogen oxides, and particulate matter, its operating costs are vulnerable to international oil prices, and it has a limited economic lifespan, requiring operators to make provisions for its future decommissioning, repowering or conversion.

1.2 Objectives of the Industrial Training

The objectives were established around the fundamental purpose of industrial attachment: converting theoretical engineering knowledge into practical understanding while developing professional conduct.

Personal learning objectives

1. To understand the operating principle and major auxiliary systems of a large reciprocating-engine power plant.
2. To observe how thermodynamics, heat transfer, fluid mechanics, internal-combustion engines, and machine design principles are implemented in an industrial environment.
3. To develop practical familiarity with fuel treatment, lubrication, compressed-air systems, cooling-water circuits, exhaust-gas systems, and power evacuation.
4. To understand the interaction between mechanical equipment, instrumentation, control systems, and electrical equipment.
5. To strengthen technical communication, teamwork, workplace discipline, safety awareness, and problem-solving ability.
6. To develop a clearer professional understanding of the role of a mechanical engineer in power generation.

Company and Training-Specific Objectives

7. To observe the complete energy-conversion chain from fuel receipt and conditioning to electricity delivery.
8. To study the MAN 18V48/60TS engine and identify its principal mechanical and combustion-related components.
9. To understand the function of critical auxiliary systems required for safe engine operation.
10. To observe operation and maintenance practices and understand why preventive and condition-based attention is important for high-value rotating machinery.
11. To understand cogeneration and the recovery of otherwise rejected exhaust heat.
12. To relate plant monitoring and control practices to pressure, temperature, flow, level, and other operating parameters.

1.3 Scope of the Report

This report is presented in accordance with the official outline prescribed for the ME 4120 Industrial Training Report. Chapter 1 provides the organisational and technical context of the host plant. Chapter 2 records the training experience in detail, describing the departments visited, the roles assumed by the trainee, and the principal technical systems studied. Chapter 3 analyses the relationship between academic theory and industrial practice, evaluates the contribution made during the attachment, reflects on challenges encountered, and offers a balanced critique of the training programme itself. Chapter 4 summarises the overall experience and presents recommendations directed both to the host company and to the university and future cohorts of students.

In addition to the core chapters, the report contains the four mandatory sections required by the department: a statement of the norms, codes and guiding values of the company; an examination of stakeholder involvement and conflicting requirements; a discussion of the ethical codes and societal duties of the company; and a workload-distribution statement confirming individual authorship. Supporting material that would normally appear in the appendices (offer letter, completion certificate, daily logbook, selected photographs and technical data sheets) is listed but not reproduced in the electronic text version.

The technical focus of the report is deliberately confined to the mechanical and process aspects of the plant. Detailed discussion of electrical protection schemes, commercial tariff structures and financial modelling lies outside the scope of a mechanical-engineering industrial-training report and is therefore omitted.

Chapter 2: Training Experience

2.1 Overview of Departments and Sections Visited

The training was structured as an academic exposure program rather than an assignment to perform unsupervised plant operation. Accordingly, the principal activities were guided observation, technical explanation, system tracing, equipment identification, and discussion with plant personnel. The presentation records exposure to the operation, maintenance, and bunker/fuel-handling sections and to the major mechanical and utility systems.

Table 2: the total task distribution of this time period.

Department / Section	Function	Approx. Time Spent
Fuel Handling / Bunker Section	Import, unloading, storage and treatment of HFO; tank-farm operation	3–4 days
Engine & Mechanical Maintenance	Engine components, mechanical hand tools, preventive maintenance	4–5 days
Lube Oil, Air & Cooling Systems	Lubrication, starting-air, instrument-air and HT/LT water systems	4 days
Co-generation / Boiler & Turbine Section	Exhaust gas boiler, steam turbine generator, water & steam generation	3 days
Electrical & Control Room (SCADA)	Alternator, transformer, switchyard and SCADA monitoring	3–4 days
Safety, HSE & General Orientation	Plant safety induction, PPE, norms and general orientation	1–2 days

Operations Department

The Operations Department is responsible for the continuous running of the generating units, the management of load schedules issued by the national load-dispatch centre, nd the real-time supervision of all process parameters through the SCADA system. Time spent in the control room allowed observation of start-up and shut-down sequences, load-following manoeuvres, alarm management and the coordination between engine operators and steam-turbine operators. Field walk-downs of the engine hall, boiler house and switchyard complemented the control-room experience and provided a three-dimensional understanding of the physical layout of the plant.

Maintenance Department

The Maintenance Department is organised into mechanical, electrical and instrumentation groups. The mechanical workshop houses a comprehensive inventory of specialised hand tools, torque equipment, lifting gear and diagnostic instruments required for the upkeep of large reciprocating engines and their auxiliaries. Mechanical Hand Tools Used in Power Plant Maintenance. Observation of routine maintenance activities, discussion of predictive-maintenance strategies, and examination of component-replacement practices formed a substantial part of the training. Particular attention was paid to the tools and procedures used for cylinder-head work, turbocharger servicing and fuel-injection equipment overhaul.

Bunker / Fuel-Handling Section

This section manages the receipt, storage, conditioning and transfer of Heavy Fuel Oil and Diesel Oil. The tank farm, associated pumping and heating systems, settling tanks, filters and booster units constitute a critical interface between external logistics and internal plant operation. Detailed study of the fuel-transfer schematic and the quality-control regime was undertaken.



Figure 2: Confidence Power & Energy PLC – Power Generation and Fuel Transportation.

Water-Treatment Plant

The water-treatment facility produces demineralised water of the quality required by the exhaust-gas boilers and the closed cooling circuits. The process train includes dual-media filtration, softening, two-pass reverse osmosis and electrodeionisation. Understanding the chemistry and the operational control of this train was an important element of the attachment.

Auxiliary Systems

Starting-air compressors and receivers, instrument-air dryers and distribution headers, lubricating-oil purification equipment, radiators and associated pumps complete the inventory of systems that support continuous generation. Each of these systems was examined in turn. Time allocation among the various areas was approximately balanced, with successive days devoted to particular systems so that a coherent picture of each could be formed before moving to the next. Daily debriefing sessions with fellow trainees and informal discussions with plant staff reinforced the observational learning.

2.2 Roles and Responsibilities

As an industrial trainee the primary role was observational and educational rather than operational. Safety regulations, insurance requirements and the complexity of the plant preclude independent operation of critical equipment by short-term visitors. Consequently the responsibilities assumed by the trainee were carefully bounded and included the following:

- Systematic study of system manuals, process-flow diagrams and piping-and-instrumentation diagrams supplied by the plant.
- Attendance at daily toolbox talks and safety briefings, and strict adherence to the personal-protective-equipment and permit-to-work regimes.
- Recording of process parameters displayed on SCADA screens and comparison of observed values with design or normal operating ranges.
- Preparation of structured daily notes that later formed the basis of the group technical presentation and of this formal report.
- Respectful technical dialogue with shift engineers, maintenance supervisors and section heads, seeking clarification of observed phenomena and operational practices.
- Participation in supervised walk-downs of the engine hall, tank farm, boiler house, water-treatment plant and switchyard.

Although the trainee did not perform hands-on maintenance or control-room operations, the depth of observation and the quality of the technical discussions were sufficient to achieve the learning objectives set out in Chapter 1. The collaborative preparation of presentation material with fellow trainees further developed the capacity for technical communication and teamwork.

2.3 Projects and Tasks Undertaken

The technical content of the attachment was organised around a series of focused tasks, each corresponding to a major plant system. The following subsections record the objective, methodology, tools or equipment examined, principal observations and any challenges noted for each task.

2.3.1 Fuel Handling, Treatment and Transfer System

Objective

To map the complete HFO supply chain from the point of international import to the engine injector, and to understand the successive quality-control steps that render residual fuel oil suitable for reliable combustion in medium-speed engines.

Methodology and Procedure

The study began with a review of the logistics chain: ocean transport from Singapore to Chittagong by mother vessels of 15 000–20 000 MT capacity; transfer by lighter vessels of 700–1 500 MT capacity to the Natakhola terminal; and final road delivery by 40-tonne tankers to the Bogura site. On-site inspection covered the tank-farm layout (5 000 MT storage tank, two 1 000 MT settling tanks, one 1 000 MT service tank, and a 500 KL Diesel Oil tank), the heating and temperature-control arrangements that maintain fuel at approximately 70 °C, the settling practice that allows water and sediments to separate over a minimum period of twenty-four hours, and the filtration stages that precede delivery to the engine booster units.

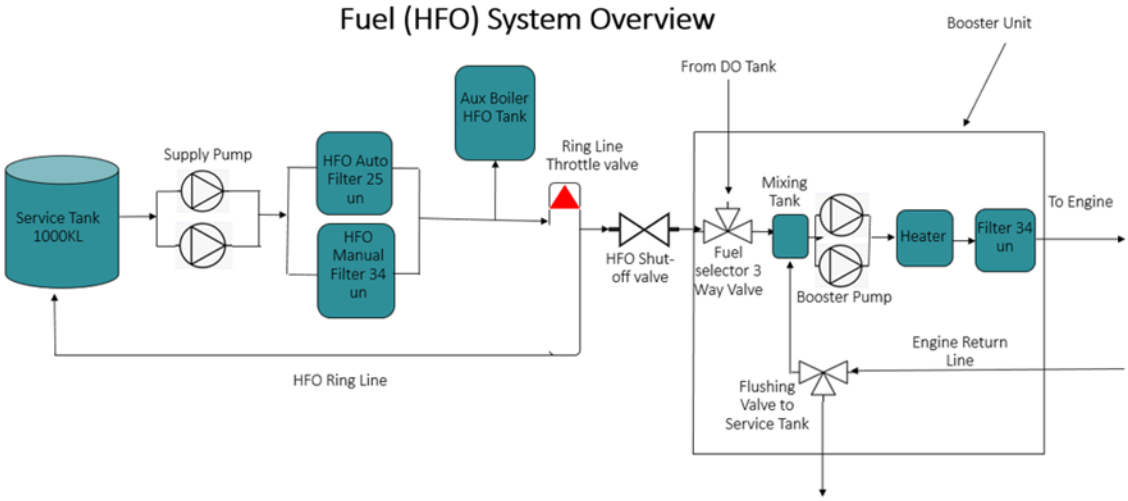


Figure 3: Heavy fuel oil (HFO) distribution and Treatment.

The fuel-transfer schematic was examined in detail. Fuel moves from the storage tank via transfer pumps into the buffer/settling tanks. After settling and heating, fuel is transferred to the service tank. From the service tank, supply pumps deliver fuel through automatic and manual filters into the booster unit, which further conditions the fuel (heating, fine filtration and pressure regulation) before it enters the engine fuel gallery and ultimately the injectors. Return lines and flushing arrangements complete the circuit and allow residual fuel to be returned to the service tank when an engine is taken out of service.

Tools and Equipment Observed

Screw unloading pumps, centrifugal transfer pumps, steam or thermal-oil heaters, centrifugal separators, automatic self-cleaning filters, booster filters, pressure and temperature transmitters, tank-level gauges, and the SCADA interface that displays tank inventories and temperatures in real time.

Results and Observations

The multi-stage heating, settling and filtration train is indispensable because residual HFO contains variable amounts of water, sediment and high-viscosity fractions. Inadequate treatment would lead to injector fouling, accelerated cylinder wear and unstable combustion. Continuous temperature control at approximately 70 °C maintains pumpability and ensures that the fuel reaches the injectors at a viscosity suitable for proper atomisation. The vertically integrated logistics chain managed by the Confidence Group—import, lighterage, road transport and on-site storage—significantly reduces exposure to external supply disruptions and gives the plant a degree of operational resilience that is valuable in a region with limited alternative fuel infrastructure.

Challenges and Mitigations

Viscosity varies with ambient temperature and with the particular batch of fuel received; water ingress can occur during transfer operations. The plant mitigates these risks through extended settling times, continuous heating, automatic differential-pressure monitoring of filters, and laboratory checks of fuel quality. Operators remain vigilant for any deviation from normal temperature or pressure trends that might indicate a developing problem in the fuel system.

2.3.2 MAN 18V48/60TS Engine Architecture and Performance

Objective

To develop a working understanding of the mechanical architecture, combustion system, turbocharging arrangement and performance characteristics of the principal generating engines installed at the plant.

Methodology

Guided inspection of the engines in the engine hall was combined with study of manufacturer literature and plant documentation. Principal structural and combustion components examined included the cylinder crankcase, cylinder heads, pistons, connecting rods, crankshaft, camshaft, rocker-arm casing, valves, fuel-cam followers, fuel-injection valves, turbochargers and flywheel. Particular attention was paid to the two-stage turbocharging system (high-pressure and low-pressure turbochargers with intercooling) that characterises the 48/60TS series.

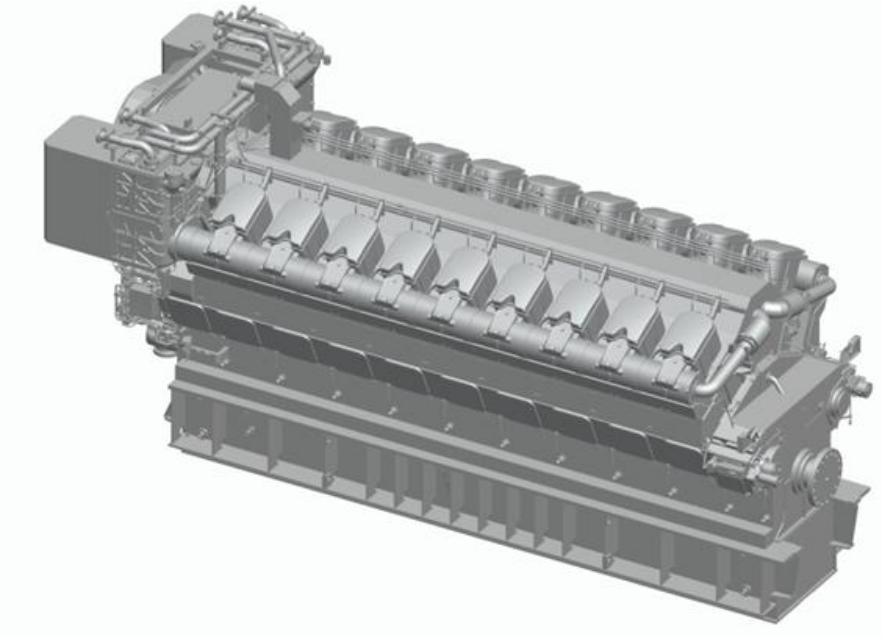


Figure 4: MAN 18V48/60TS Engine Architecture.

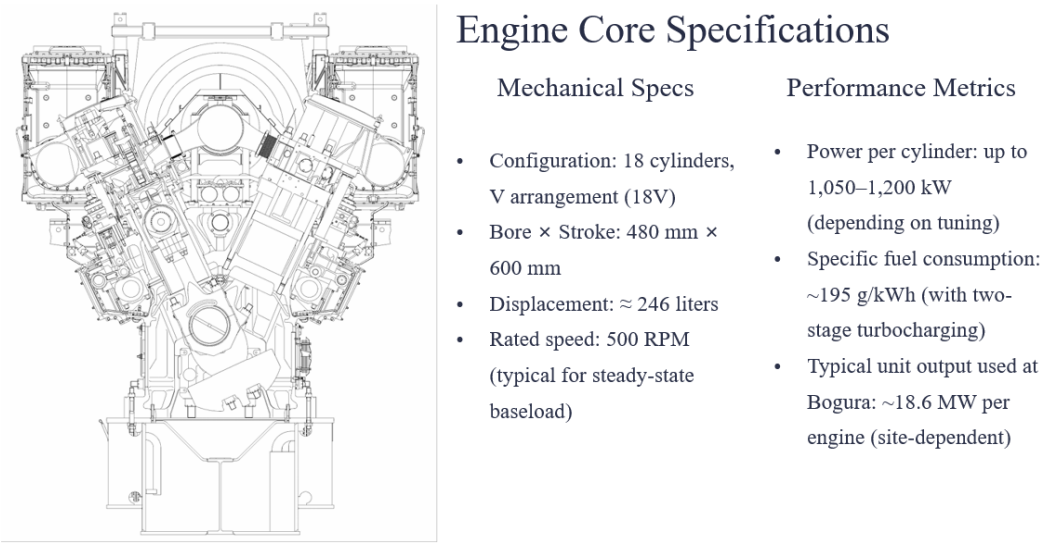


Figure 5: Cross-section view of the engine.

Observations on Design and Operation

The two-stage turbocharging arrangement, combined with charge-air cooling, allows the engine to achieve high mean effective pressures while controlling thermal loading of the combustion chamber components. The relatively low rotational speed reduces mechanical stress and contributes to long component life when the engine is properly maintained and lubricated. Direct observation of the engines under load, together with discussion of start-up and loading procedures with shift engineers, provided a concrete illustration of the relationship between theoretical engine cycles and the practical behaviour of a large industrial machine.

Mechanical Hand Tools

The maintenance workshop contains an extensive inventory of specialised tools required for engine work. Among those observed were ring and combination spanners, D-shackles, try-squares, air guns, hand blowers, impact guns, rubber and ball-pein hammers, heavy-duty ratchets, T-handles, ferrule spanners, L-keys, sockets, adapters and eye-bolts. The correct selection and use of these tools, together with the application of specified torque values, form an essential part of the maintenance culture of the plant.



Figure 6: Mechanical Hand Tools.

2.3.3 Lubricating Oil System

Objective

To understand the design and function of the closed-loop lubricating-oil system that protects the moving parts of the MAN engines.

Purpose of the System

The lubricating-oil system performs five essential functions: reduction of friction between moving surfaces; removal of heat generated by friction and combustion; cleaning of contaminants and wear particles; prevention of corrosion; and assurance of smooth, reliable engine operation. Proper lubrication is therefore fundamental to engine longevity, efficiency and availability.

System Description

Each engine is equipped with a lubricating-oil sump of substantial capacity (approximately 23 m^3). An engine-driven pump and a separate priming pump circulate oil through coolers, automatic back-flush filters and a commissioning filter before the oil is distributed to the engine's moving parts, including the main bearings, connecting-rod bearings, camshaft, and cylinder-liner lubrication points.

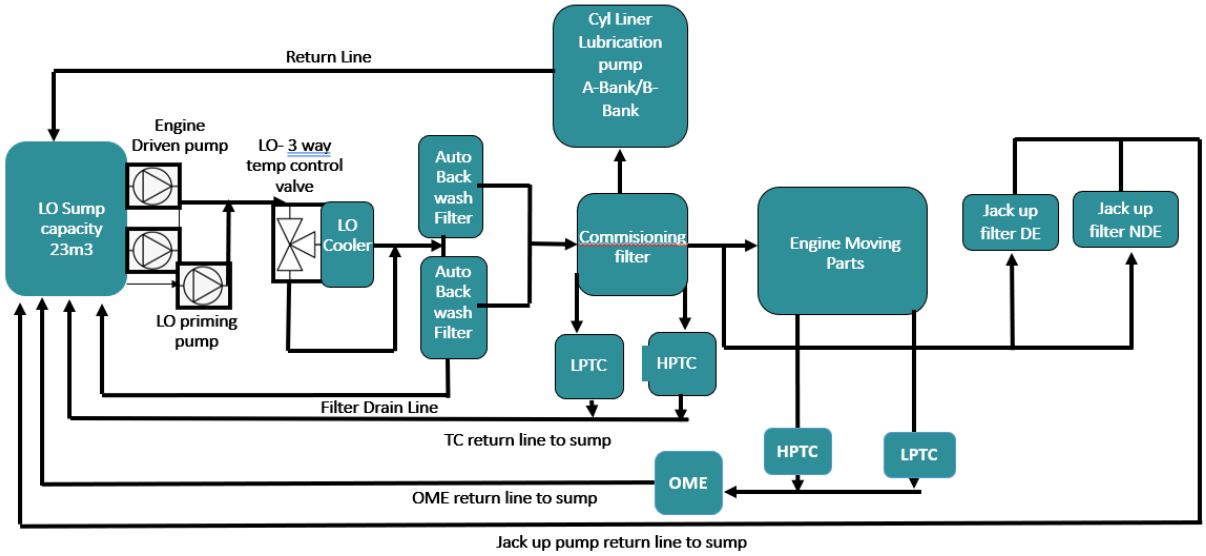


Figure 7: Lube oil system overview.

Separate jack-up filters serve the drive-end and non-drive-end of the engine. Return lines bring oil back to the sump, closing the circuit. A parallel transfer and purification system allows fresh oil to be received into a dedicated storage tank, used oil to be drained to a used-oil tank, and continuous or batch purification to be performed by centrifugal separators.

Observations

The advantages of a well-maintained lubrication regime—longer engine life, reduced maintenance cost, improved efficiency, lower mechanical wear and stable performance—were repeatedly emphasised by maintenance personnel. Continuous monitoring of oil pressure, temperature and differential pressure across filters provides early warning of developing problems. The presence of both engine-driven and electrically driven pumps ensures that lubrication can be maintained during start-up, shut-down and emergency conditions.

2.3.4 Cooling Water Systems (HT and LT Circuits)

Objective

To examine the dual high-temperature and low-temperature cooling-water circuits that manage the thermal balance of the engines and their auxiliaries.

High-Temperature Circuit

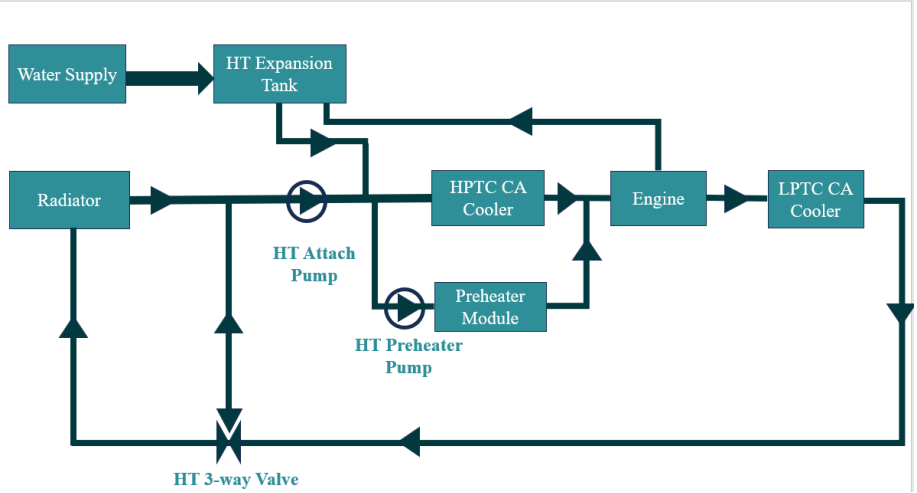


Figure 8: High-Temperature Water Circuit

The HT circuit cools the engine block, cylinder heads and cylinder liners. It is designed to operate at a relatively high temperature (approximately 70–90 °C) so that the combustion-chamber surfaces remain warm enough to support efficient combustion and to minimise corrosive condensation of combustion products. The circuit includes an expansion tank, radiators or heat exchangers, a three-way temperature-control valve, an engine-driven or attached circulating pump, and a pre-heater module that raises water temperature before start-up.

Low-Temperature Circuit

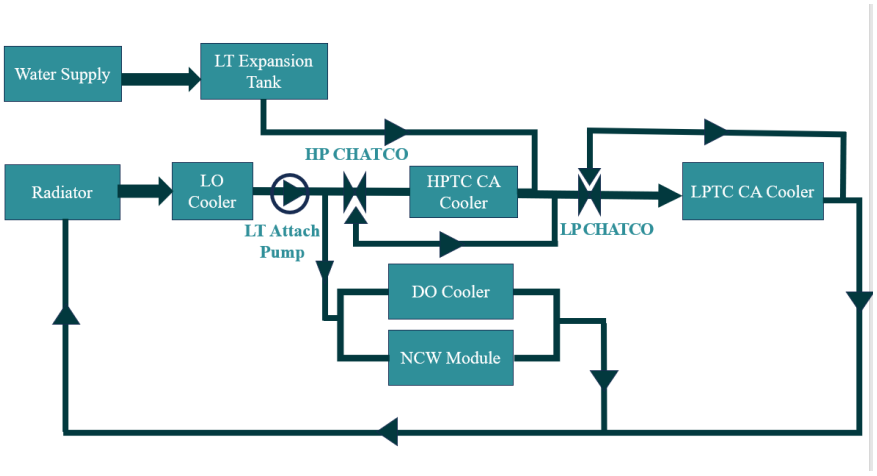


Figure 9: Low-Temperature Water Circuit

The LT circuit operates at a lower temperature (approximately 60 °C) and cools a range of auxiliary components: the lubricating-oil cooler, the high-pressure and low-pressure charge-air coolers, the fuel-nozzle coolers, the booster unit and other secondary heat loads. Because these components require effective heat rejection at moderate temperature levels, a separate circuit is provided. The LT circuit likewise includes an expansion tank, radiators, circulating pumps and appropriate control valves.

Observations

The separation of HT and LT duties is a deliberate design choice that optimises both combustion efficiency and component cooling. Observation of the circuits under operating conditions, together with examination of the associated P&IDs, clarified the flow paths and the role of the three-way valves in maintaining set-point temperatures. The importance of water chemistry control (to prevent scale, corrosion and biological fouling) was also noted.

2.3.5 Starting Air and Instrument Air Systems

Starting-Air System

Large medium-speed engines are started by compressed air. The starting-air system at Bogura comprises three compressors (SAC 01, 02 and 03) that charge two receivers, each of 5 000 litre capacity and design pressure 30 bar. Starting air is supplied to the engines at pressures up to 30 bar; control air and pilot air operate at lower pressures (down to approximately 6–8 bar depending on the particular function). The system also supports a slow-turn device used during maintenance. Observation of the compressors, receivers and distribution piping, together with discussion of start-up sequences, illustrated the critical role of a reliable high-pressure air supply.

Instrument-Air System

The instrument-air system supplies clean, dry compressed air to pneumatic control valves, actuators, analysers and safety devices throughout the plant. Because many automatic control loops depend on air pressure, the instrument-air system is regarded as a critical utility. The arrangement includes instrument-air compressors, dryers, a receiver (maximum working pressure typically 16 bar) and a distribution header that serves the engine hall, auxiliaries, boilers and other consumers. Moisture or oil contamination of instrument air can cause control-valve malfunctions

and safety-system failures; consequently the drying and filtration stages are subject to careful maintenance.

2.3.6 Water Treatment and Demineralised Water Production

Objective

To understand the process train that produces demineralised water of the quality required by the exhaust-gas boilers and the closed cooling circuits.

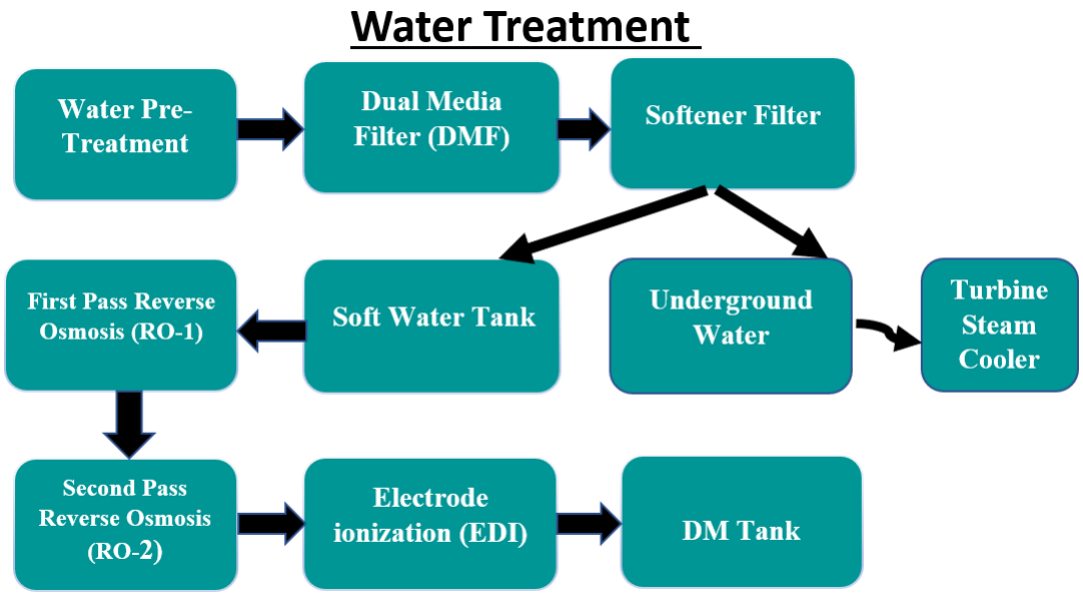


Figure 10: Raw water treatment process of

Process Description

Raw water first undergoes pre-treatment and then passes through a dual-media filter (anthracite and sand) that removes suspended solids, sediments and turbidity. The filtered water proceeds to a softener that reduces hardness by ion exchange. Softened water is stored and then fed to a two-pass reverse-osmosis system. The RO permeate is further polished by electrodeionisation to produce demineralised water of the required conductivity and silica content. The finished DM water is stored in a dedicated tank (capacity approximately 100 MT) from which it is drawn for boiler feed and for make-up to the closed cooling circuits.

Observations

The multi-stage treatment train reflects the stringent water-quality requirements of high-pressure boilers and of closed cooling systems. Continuous monitoring of conductivity, pH and silica, together with periodic laboratory analysis, ensures that the product water remains within specification. The presence of both RO and EDI stages provides a robust polishing capability and reduces the chemical consumption associated with older mixed-bed ion-exchange systems.

2.3.7 Exhaust Gas System and Co-Generation Train

Objective

To examine the recovery of residual exhaust energy and its conversion into additional electrical output and process heat.

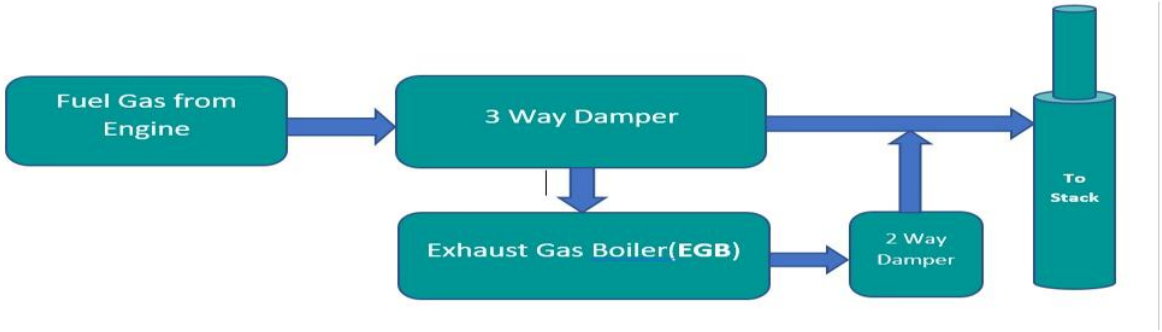


Figure 11: Co-Generation System.

Exhaust-Gas Path

Hot exhaust gas leaves each engine at temperatures typically in the range 550–565 °C under load. The gas first passes through the high-pressure turbocharger and then the low-pressure turbocharger, with a waste-gate available to bypass the turbines under certain operating conditions. After the turbochargers the gas enters the exhaust-gas boilers. A three-way damper arrangement allows the gas either to be directed through the boilers or, when necessary, to be sent directly to the stack.

Steam Generation and Utilisation

Demineralised water is deaerated in high-pressure and low-pressure deaerators to remove dissolved gases that would otherwise cause corrosion. The deaerated water is fed to the exhaust-gas boilers, which generate steam. High-pressure steam drives the steam-turbine generators (each approximately 3.6 MW), whose output is synchronised and exported through the plant's electrical system. Low-pressure steam is supplied to a header that serves engine-auxiliary heating duties—principally the heating of fuel and lubricating oil. Condensate is returned to the deaerators, closing the water–steam cycle and minimising the demand for fresh make-up water

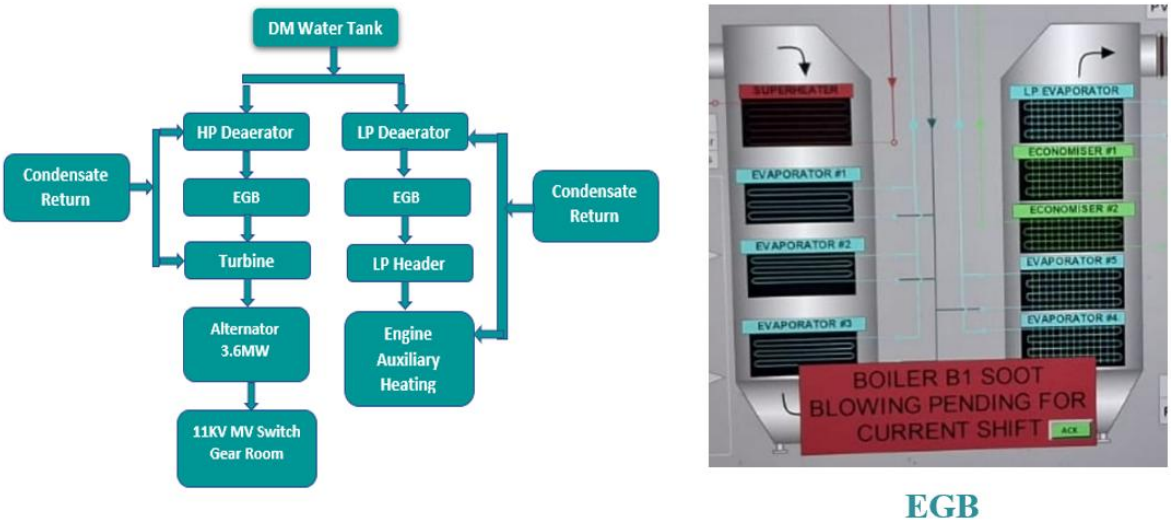


Figure 12: Boiler & turbine process overview

Observations

The co-generation arrangement raises the overall thermal efficiency of the plant by converting a substantial fraction of the residual exhaust energy into useful work and process heat. The closed condensate return further improves efficiency and reduces water consumption. Observation of the boilers, turbines and associated piping, together with discussion of operating procedures, provided a concrete illustration of the Rankine cycle applied to waste-heat recovery in a reciprocating-engine context.

2.3.8 Electrical Power Evacuation and SCADA Overview

Power Evacuation Path

Mechanical energy produced by each engine is converted into electrical energy by a directly coupled synchronous alternator. Typical output voltage is 11 kV at 50 Hz. Before the generator can be connected to the grid, voltage, frequency, phase sequence and phase angle must be matched; this synchronisation is performed at the synchronisation panel. A generator circuit breaker protects the machine against short-circuit, over-current and other faults. The generator voltage is then stepped up by a power transformer to 132 kV for transmission. The high-voltage switchyard contains circuit breakers, isolators, lightning arresters, current and potential transformers, and busbars that form the final interface with the national grid.



Figure 13: Electricity production to transmission national grid

SCADA System

The Supervisory Control and Data Acquisition system constitutes the central nervous system of the plant. It continuously monitors pressures, temperatures, flows, levels, speeds and equipment status throughout the facility. Operators interact with the process through human–machine interface screens that display both real-time values and historical trends. Automatic control loops

maintain set-points for critical parameters, while alarm management draws attention to deviations

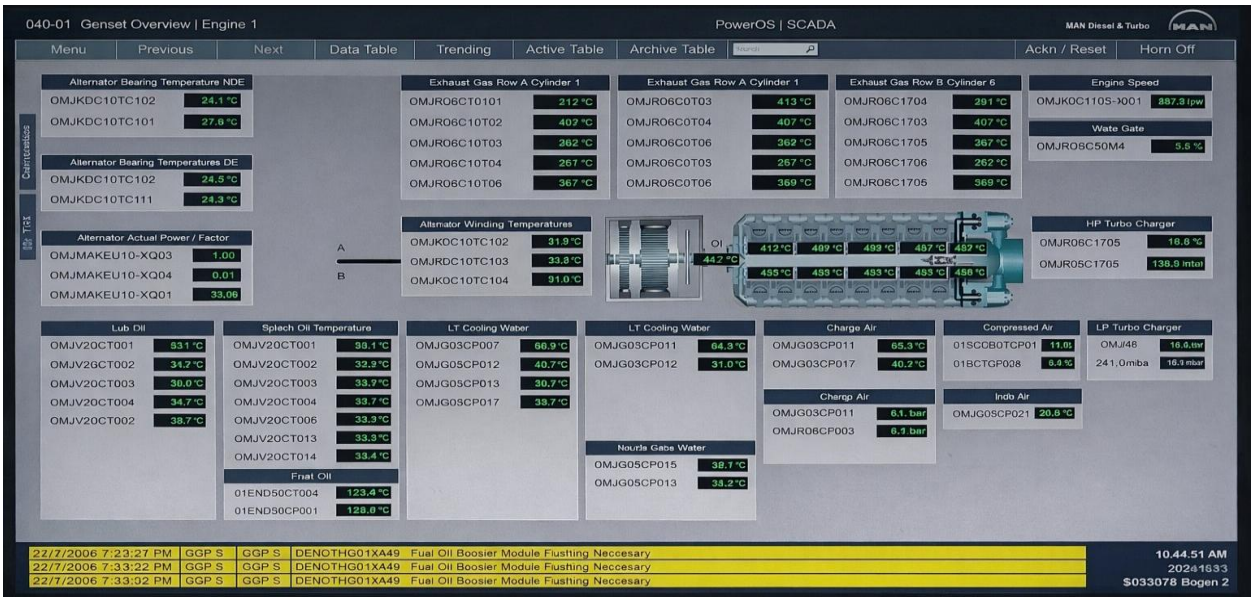


Figure 14: SCADA system of the Engine room.

The SCADA system also records events for later analysis and supports both local and, where configured, remote supervision. Observation of the SCADA screens during normal operation and during simulated or actual load changes illustrated the depth of instrumentation and the degree of automation that characterise a modern IPP plant.

2.4 Skills Gained and Enhanced

Technical Skills

- Ability to interpret process-flow diagrams, piping-and-instrumentation diagrams and SCADA displays for a complex thermal power plant.
- Working knowledge of the architecture, turbocharging and performance characteristics of large medium-speed diesel engines of the MAN 18V48/60TS class.
- Understanding of multi-stage fuel treatment, closed-loop lubrication, dual-circuit cooling, starting-air and instrument-air systems, and exhaust-gas heat recovery.
- Familiarity with industrial water-treatment processes (filtration, softening, reverse osmosis and electrodeionisation) applied to power-plant duty.

- Awareness of the specialised hand tools, torque practices and safety protocols employed in the maintenance of heavy reciprocating engines.
- Appreciation of the logistical, contractual and environmental constraints that shape the operation of an HFO-fired IPP in Bangladesh.

Soft Skills

- Professional communication with engineers, supervisors and fellow trainees in a formal industrial setting.
- Teamwork in the collaborative preparation of technical presentation material and daily observational records.
- Heightened awareness of industrial safety culture, personal protective equipment requirements and permit-to-work procedures.
- Improved capacity for structured observation, note-taking and the subsequent synthesis of technical information into a coherent formal report.
- Time management under the disciplined schedule of a multi-week industrial attachment.

Chapter 3: Analysis and Discussion

3.1 Application of Theoretical Knowledge

The industrial attachment provided numerous opportunities to observe the practical expression of theoretical concepts studied in the mechanical-engineering curriculum. Thermodynamics and heat-transfer principles were directly visible in the exhaust-gas boilers and the dual HT/LT cooling circuits. The Rankine cycle, previously examined in idealised textbook form, was observed in the real steam-turbine train that recovers residual heat from the engine exhaust. The improvement in overall plant efficiency achieved by co-generation illustrated the economic and thermodynamic value of waste-heat recovery.

Fluid-mechanics concepts underpinned the understanding of fuel and lubricating-oil pumping, the pressure drops that occur across filters, and the flow of charge air through the turbochargers and intercoolers. The design of the tank-farm transfer system and the booster-unit circuit demonstrated

the practical application of pump-selection criteria, pipe-sizing considerations and the management of viscous fluids at elevated temperature.

Internal-combustion-engine theory found concrete expression in the MAN 18V48/60TS units. The relationship between mean effective pressure, specific fuel consumption and turbocharging efficiency, the thermal loading of combustion-chamber components, and the role of injection timing and fuel atomisation quality were all illuminated by observation of the engines under load and by discussion with operating staff. The relatively low rotational speed of 500 rpm, chosen for base-load durability and direct coupling to a 50 Hz alternator, provided a practical illustration of the design compromises that characterise large industrial engines.

At the same time, the attachment revealed the inevitable discrepancies between idealised textbook assumptions and industrial reality. Fuel quality varies from batch to batch; ambient temperature affects viscosity and therefore pump and injector performance; heat exchangers foul over time; and sensors drift. These real-world phenomena underscore the importance of continuous monitoring, disciplined maintenance and engineering judgement that goes beyond pure calculation. The experience therefore reinforced both the power of theoretical frameworks and the necessity of adapting those frameworks to the messy conditions of actual plant operation.

3.2 Contribution to the Company

Given the short duration of the attachment and the safety restrictions that properly limit the activities of industrial trainees, the direct operational contribution of any individual trainee is necessarily modest. The trainee did not perform independent maintenance, operate critical equipment, or assume responsibility for generation schedules. Nevertheless, several forms of contribution can be identified.

First, the systematic documentation of process descriptions, the preparation of presentation material, and the careful recording of observations provided a modest knowledge-capture benefit. Material prepared by the trainee group was reviewed by plant staff and may serve as a reference for future cohorts of visitors. Second, the presence of motivated engineering students constitutes an informal demonstration of the company’s willingness to engage with academic institutions—an intangible but positive contribution to corporate reputation and to the broader engineering education ecosystem. Third, the questions posed by trainees occasionally prompted experienced

staff to articulate practices that might otherwise remain tacit, thereby reinforcing the plant's own reflective capacity.

In quantitative terms the contribution remains small; in qualitative terms it forms part of the reciprocal relationship that ought to exist between industry and the universities that supply its future engineers.

3.3 Challenges Encountered and Learning Points

Technical Challenges

The sheer complexity of the multi-system interactions—fuel, air, water, steam, lubrication, control and electrical—presented a steep learning curve. Assimilating the function of each system, the interfaces between systems, and the operating philosophy that governs the whole plant required sustained concentration and repeated clarification. The limited duration of the attachment constrained the depth of exposure that could be achieved, particularly with respect to major overhaul activities that occur only at long intervals.

Interpersonal and Organisational Challenges

Adapting to the hierarchy, communication style and time discipline of an industrial organisation required conscious adjustment. Shift patterns, toolbox talks and the formal permit-to-work culture differ markedly from the more informal environment of a university laboratory. The need to balance curiosity with respect for operational priorities and safety protocols was a constant consideration.

Responses and Learning Points

These challenges were addressed through disciplined daily note-taking, systematic study of the diagrams and manuals supplied by the plant, repeated clarification with supervisors, and collaborative discussion among the trainee group. The principal learning points that emerged are the following. First, successful plant operation depends as much on disciplined procedures, continuous monitoring and a strong safety culture as on the excellence of the hardware itself. Second, the integration of multiple auxiliary systems is as critical to availability as the performance

of the main engines. Third, the contractual, logistical and environmental context of an IPP shapes every operational decision; technical excellence alone is not sufficient. Fourth, the ability to communicate clearly with experienced practitioners, to listen carefully, and to synthesise observations into coherent technical documentation is itself a professional skill of high value.

3.4 Strengths and Weaknesses of the Training Programme

Strengths

The host plant provided access to a modern, well-maintained combined-cycle facility of substantial capacity. Staff at all levels proved knowledgeable and cooperative, willing to explain systems and to answer questions within the limits of operational constraints. The twenty-day duration, while not excessive, was sufficient to allow systematic coverage of the major mechanical systems. The combination of control-room observation, field walk-downs and guided discussion produced a balanced learning experience. The requirement to prepare a formal technical presentation further reinforced the educational value of the attachment.

Weaknesses and Areas for Improvement

Hands-on involvement was necessarily limited by safety and insurance considerations; future programmes might usefully incorporate carefully supervised practical exercises on non-critical equipment or on training simulators if available. Formal performance-assessment metrics for the trainee were not prominent; the introduction of a short structured project (for example, an efficiency calculation based on observed data, or a simple failure-mode-and-effects analysis of a selected subsystem) would strengthen the educational outcome. Exposure to electrical protection schemes and to the commercial aspects of the Power Purchase Agreement remained relatively limited; a brief overview session on these topics would provide useful context. Finally, a structured closing debrief involving both plant supervisors and university faculty would help to consolidate learning and to identify improvements for subsequent cohorts.

Overall, the strengths of the programme substantially outweigh its limitations. With modest enhancements of the kind suggested above, the industrial-attachment experience at Confidence

Power Bogura could become an even more effective bridge between academic study and professional practice.

Chapter 4: Conclusion and Recommendations

4.1 Conclusion

The twenty-day attachment at Confidence Power & Energy PLC, Bogura, was fruitful and achieved the set objectives. I gained a comprehensive understanding of a 113 MW HFO-based combined-cycle independent power plant, acquired knowledge of MAN 18V48/60TS engines and their auxiliaries, and saw how thermodynamic, fluid-mechanical, and power-plant engineering principles are applied in practice. The training also helped me realize the technical and non-technical aspects of engines and their auxiliaries used in modern medium-speed plants. The plant plays a significant role in the power supply network in Bangladesh's northern region. However, the high emission-producing fuel, expensive fuel prices, and fifteen-year contract hinder the plant's potential to become one of the best in the region. The soft skills learned in the training process will also help me improve my communication, teamwork, safety knowledge, and documentation skills in my professional career.

In conclusion, the training was an eye-opener to my academic knowledge about thermal power plants. I have gained new knowledge and skills that will benefit my future engineering career.

4.2 Recommendations

4.2.1 Recommendations for the Company

- Continue and, where resources permit, formalise the structured induction and technical-orientation programme for engineering trainees. The inclusion of a short, measurable project component (for example, an efficiency or heat-balance calculation based on plant data, or a focused reliability study of a selected subsystem) would further enhance the educational return for both the trainee and the company.
- Maintain the existing rigorous regime of fuel-quality monitoring, filter maintenance and temperature control. Given the sensitivity of residual-fuel engines to fuel contaminants, continued investment in laboratory capability and in operator training remains justified.

- Continue to monitor developments in emission-control technology and in national environmental regulation. Incremental improvements in the management of sulphur oxides, nitrogen oxides and particulate matter will protect both regulatory compliance and the company's long-term social licence to operate.
- Initiate early strategic planning for the post-2034 period. Options may include life-extension assessments, repowering with cleaner fuels, conversion to dual-fuel or gas operation if pipeline infrastructure becomes available, or eventual site redevelopment. Early planning reduces the risk of stranded assets and allows orderly transition.
- Strengthen the systematic capture of lessons learned from both routine maintenance and forced outages. Documented case studies of equipment failures, root-cause analyses and corrective actions would constitute a valuable resource for future operators and for subsequent cohorts of trainees.
- Consider the periodic organisation of technical seminars or short courses open to university faculty and advanced students. Such engagement reinforces the company's contribution to the national engineering-education ecosystem and helps to identify high-potential graduates for future recruitment.

4.2.2 Recommendations for the University and Future Students

- Provide a structured pre-attachment briefing package that includes typical process-flow diagrams, a glossary of plant-specific terminology, an overview of industrial safety protocols, and a short reading list on medium-speed engine technology and HFO logistics. Students who arrive with a basic conceptual map of the plant progress more rapidly once on site.
- Encourage students to prepare targeted technical questions in advance of the attachment and to maintain a disciplined daily log that records observations, parameters, questions and reflections. The quality of the final report is strongly correlated with the quality of the contemporaneous notes.
- Explore the possibility of extending the duration of future attachments, or of arranging follow-up visits during major planned outages, so that students can observe overhaul practices that are not visible during normal running periods.

- Institute a short post-attachment reflective seminar at which students who have trained at different host organisations can share observations and compare experiences. Such a seminar would broaden the educational benefit beyond the individual plant visited.
- Consider the introduction of a modest formal assessment component (for example, a technical presentation evaluated jointly by plant and university supervisors, or a short analytical assignment based on data collected during the attachment) so that the learning outcomes of the industrial-training course are made more explicit and measurable.
- Maintain and, where possible, expand the network of industrial partners so that future cohorts continue to have access to high-quality training opportunities across a range of mechanical-engineering domains.

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Norms, Codes and Guiding Values of the Company

Confidence Power & Energy PLC works in a framework of statutory, contractual, and internal constraints that govern the operation of the plant. Statutory requirements include the Power Purchase Agreement, the Bangladesh Power Development Board regulations, the Department of Environment rules, and factory Acts, labor laws, and other enactments relating to employment and safety. In addition, the company has internal behavioral norms and goals relating to the production process. For example, a key value is operational excellence, which is manifested in the ability to operate each generating unit at full power and achieve high levels of availability. Supply of electricity is viewed as an obligation to the customers rather than an option, and the company strives to be reliable and consistent in its operations. The company values safety, and this is evident in the provision of personal protective equipment in appropriate work areas, holding toolbox talks on job safety, and the requirement of permits to work on specialized tasks. Another key organizational norm is the adherence to procedures, including those for start-up, shutdown, load changes, fuel supply, maintenance, and other aspects of operations. These procedures are mandatory and should only be performed in the specified manner. The SCADA system enables continuous monitoring of the plant's operations to ensure that they conform to procedures. In

general, the company believes that processes must always be improved based on the evaluation of deviations that may occur or have occurred. As such, there are continual investigations of operating practices, accidents, and near-accidents, which are used to update and improve the procedures, manuals, and training materials.

Finally, there is respect for the hierarchy, though junior workers can engage in open discussions with senior staff, including shift engineers, on technical matters. The company's norms and values enable it to operate reliably and consistently while also promoting professional growth for its employees.

Extent of Stakeholder Involvement and Conflicting Requirements

Confidence Power plant at Bogura – Interests of stakeholders:

The operation of an Independent Power Producer plant such as Confidence Power involves numerous stakeholders, each of whom has some level of responsibility and interest. Some of the stakeholders are shareholders, host country entities, international oil suppliers and transporters, employees, local communities and authorities, and regulatory agencies. The interrelations between these stakeholders are not always straightforward. The manager should be aware of the potential conflicts of interest between different stakeholders.

First, Bangladesh Power Development Board (BPDB) is the offtaker under the Power Purchase Agreement (PPA). It is interested in receiving the declared capacity at specified technical parameters and at an affordable price. The PPA defines availability targets and penalties in case of a shortfall. Therefore, the plant's operational focus is on maintaining the highest possible availability. At the same time, the management is concerned with high oil prices, particularly due to the costs that may arise from the fuel price adjustment clause. Thus, there is a potential conflict between BPDB and plant management in terms of availability and fuel costs.

Next, the management of the Confidence Group is interested in maintaining adequate revenue, considering the large initial investments. This is closely related to the concerns about the operating costs; as the manager mentions, they have had to limit the costs due to the pressure from the shareholders. Therefore, there may be a conflict with the employees in terms of maintenance, spare parts, and emission abatement equipment. Similarly, the employees' primary concern is safety, which is consistent with the interests of BPDB, the host country, and the regulatory agencies. The need to limit costs may create a conflict by reducing the safety margin.

Furnace oil suppliers may have an interest in the size of the purchase orders and the promptness of payments, which can create a conflict with the plant's own budgetary constraints. In addition, local environmentalists are likely to raise concerns about the emissions and the waste water from the plant. Finally, the regulatory agencies should ensure that emission standards, safety, and working conditions are met. Meanwhile, the plants and their stakeholders are concerned with cost efficiency, which is most evident in the concerns about the PPA costs. As mentioned, the PPA has a life span of 15 years, which means that the company must consider the end-of-contract options. Thus, there is a trade-off between the ability to recover costs and the need to plan for the future.

To conclude, the manager should be mindful of the interests of the stakeholders and the potential conflicts that he may encounter at work. He should continue to work closely with BPDB to address their concerns. In addition, the manager will need to balance the concerns regarding operating costs with the need to guarantee safety and availability. This will require him to maintain the communication channels with local authorities and environmentalists while also planning for the future. The technical staff of the plant should be aware of the need to consider the bigger picture while also being mindful of the immediate concerns.

Ethical Codes and Societal Duties of the Company

As a licensed Independent Power Producer, Confidence Power & Energy PLC bears a set of ethical obligations and societal duties that extend beyond the purely commercial terms of the Power Purchase Agreement. The most fundamental of these duties is the reliable supply of electricity to the northern region of Bangladesh. Access to continuous power underpins industrial activity, agricultural processing, commercial enterprise, education, healthcare and household welfare. In a region that has historically experienced supply shortfalls, the contribution of the Bogura plant to grid stability and energy access constitutes a tangible public good.

Ethical conduct in the commercial sphere requires truthful and accurate reporting of generation data, availability figures and technical performance parameters to BPDB. Misrepresentation of these data would undermine the integrity of the contractual relationship and the fairness of the tariff regime. Transparent handling of fuel procurement, logistics contracts and financial transactions is equally required; any form of corruption or collusion would violate both legal and ethical standards.

Fair labour practices form another core ethical obligation. Employees are entitled to safe working conditions, fair remuneration, reasonable working hours and freedom from discrimination or harassment. The plant's visible emphasis on safety culture and personal protective equipment is consistent with this obligation; continued vigilance is required to ensure that commercial pressures never compromise the physical well-being of the workforce.

Environmental responsibility presents a more complex ethical challenge. Residual fuel oil combustion produces higher emissions of sulphur oxides, nitrogen oxides, particulate matter and greenhouse gases than natural-gas or renewable alternatives. While the plant operates within the environmental standards currently applicable in Bangladesh, the ethical duty of a responsible operator includes continuous monitoring of emissions, exploration of abatement options that are technically and economically feasible, and participation in the longer-term national transition toward a cleaner energy mix. The finite contractual life of the plant creates both a risk of stranded high-emission assets and an opportunity to plan for cleaner successor technologies.

Finally, the company has a societal duty to contribute to the development of national engineering capacity. The willingness to host industrial trainees, to share technical knowledge, and to engage with universities forms part of this broader responsibility. In fulfilling these ethical and societal duties, the company aligns its commercial performance with the longer-term interests of the communities and the nation it serves.

Workload Distribution Statement

The industrial training report was prepared as a group activity. The names and roll numbers should be inserted according to the actual contribution of each student. The following table provides a clear format for declaring the distribution of work.

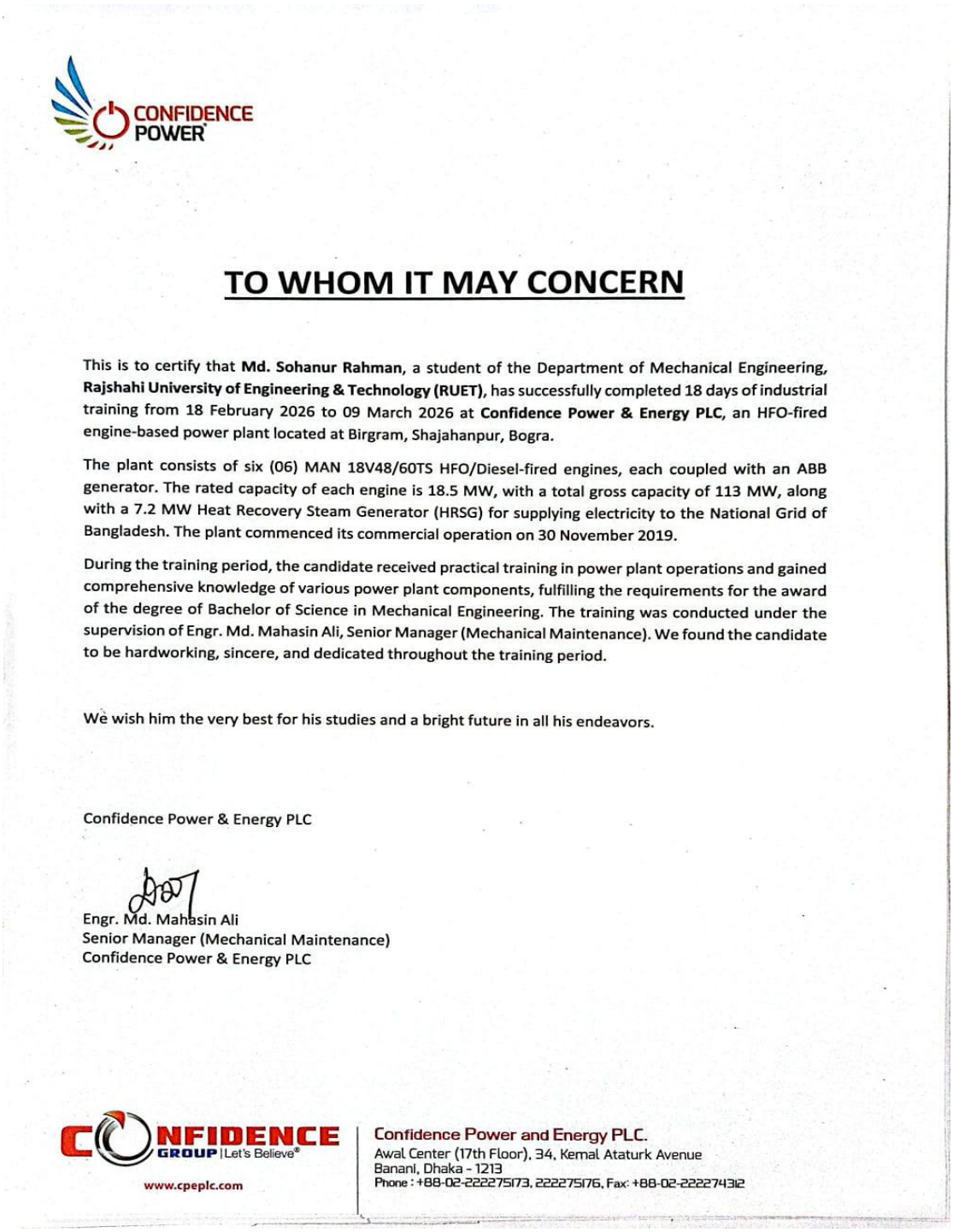
Student	**Roll No.**	**Assigned section/work**
Md. Ashikur Rahman Md. Sohanur Rahman Bahktiar Ahmed Sajib	2102018 2102026 2102036	Introduction, company profile, training experience
Mir Sajib Ajmain Hasnat Alvi Shayekh Mohiuddin Mashuq	2102046 2102048 2102051	Analysis and discussion
Jiaur Rahman Jewel Amit Kumar Sarker	2102052 2102054	Conclusion, recommendations and appendices
All members	All of above	Review, editing

APPENDIX A: TRAINING LOG SUMMARY

Date	Main activity / learning focus
18-02-2026	Plant orientation and safety briefing
19-02-2026	Plant layout and engine-generator familiarization
20-02-2026	Engine construction and major components
21-02-2026	Engine operating principles / observation
22-02-2026	Fuel system and injection arrangement
23-02-2026	Cooling-water system
24-02-2026	Lubricating-oil system
25-02-2026	Starting-air system
26-02-2026	Control and monitoring
27-02-2026	Maintenance practices / inspection
28-02-2026	Review and technical discussion
01-03-2026	Engine auxiliary systems
02-03-2026	Operational observations
03-03-2026	Maintenance/condition monitoring
04-03-2026	Data/diagram review
05-03-2026	Safety and environmental practices
06-03-2026	Consolidation of technical notes
07-03-2026	Group discussion and presentation preparation
08-03-2026	Final review of attachment learning
09-03-2026	Completion/final briefing

Industrial Attachment Certificate

Certificate of Completion for Industrial Training at Confidence Power & Energy PLC, Bogura Power Plant.



CONFIDENCE POWER

TO WHOM IT MAY CONCERN

This is to certify that **Md. Sohanur Rahman**, a student of the Department of Mechanical Engineering, **Rajshahi University of Engineering & Technology (RUET)**, has successfully completed 18 days of industrial training from 18 February 2026 to 09 March 2026 at **Confidence Power & Energy PLC**, an HFO-fired engine-based power plant located at Birgram, Shajahanpur, Bogra.

The plant consists of six (06) MAN 18V48/60TS HFO/Diesel-fired engines, each coupled with an ABB generator. The rated capacity of each engine is 18.5 MW, with a total gross capacity of 113 MW, along with a 7.2 MW Heat Recovery Steam Generator (HRSG) for supplying electricity to the National Grid of Bangladesh. The plant commenced its commercial operation on 30 November 2019.

During the training period, the candidate received practical training in power plant operations and gained comprehensive knowledge of various power plant components, fulfilling the requirements for the award of the degree of Bachelor of Science in Mechanical Engineering. The training was conducted under the supervision of Engr. Md. Mahasin Ali, Senior Manager (Mechanical Maintenance). We found the candidate to be hardworking, sincere, and dedicated throughout the training period.

We wish him the very best for his studies and a bright future in all his endeavors.

Confidence Power & Energy PLC

Engr. Md. Mahasin Ali
Senior Manager (Mechanical Maintenance)
Confidence Power & Energy PLC

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Memories and Learning from Industrial Attachment





